

TRIER PETRISBERG, Germany

WMO: 106090

Lat: 49.7478N

Lon: 6.6583E

Elev: 273

StdP: 98.09

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-7.9	-14.4	1.1	-9.0	-12.2	1.4	-6.6	14.0	7.1	12.7	5.9	4.4	40	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	29.7	19.1	27.8	18.4	25.9	17.5	20.2	27.3	19.2	25.4	18.4	24.1	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.4	21.9	17.1	12.6	21.4	16.2	11.9	20.5	59.1	27.1	55.7	25.3	53.0	24.3	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.4	8.1	-11.2	33.5	3.5	1.8	-13.7	34.8	-15.8	35.9	-17.8	36.9	-20.3	38.2		
(5)			-11.5	22.9	3.6	1.1	-14.1	23.7	-16.2	24.3	-18.2	24.9	-20.8	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	1.0	1.3	5.5	8.3	13.1	16.1	18.6	18.3	14.6	10.3	4.8	2.8	(6)
(7)		DBStd	7.28	4.78	4.53	3.57	3.58	3.76	3.62	3.44	3.20	3.12	3.43	3.76	3.87	(7)
(8)		HDD10.0	1181	279	244	145	73	14	1	0	0	1	38	159	226	(8)
(9)		HDD18.3	3302	538	477	397	300	166	87	40	42	117	250	405	483	(9)
(10)		CDD10.0	1036	0	1	6	23	109	183	268	256	141	47	3	1	(10)
(11)		CDD18.3	118	0	0	0	0	3	19	49	39	7	1	0	0	(11)
(12)		CDH23.3	1077	0	0	0	1	47	168	454	352	54	1	0	0	(12)
(13)		CDH26.7	290	0	0	0	0	4	33	144	102	7	0	0	0	(13)
(14)	Wind	WSAvg	3.3	4.3	3.9	4.0	3.5	3.0	2.8	2.7	2.6	2.9	3.2	3.4	3.6	(14)
(15)	Precipitation	PrecAvg	748	64	53	53	46	69	67	76	70	61	62	71	73	(15)
(16)		PrecMax	1071	192	134	150	129	130	177	218	162	147	192	132	212	(16)
(17)		PrecMin	541	23	7	9	0	13	8	7	16	14	4	3	2	(17)
(18)		PrecStd	124	35	30	28	26	27	40	41	36	35	37	32	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	14.0	19.7	22.3	26.9	29.6	32.4	32.7	27.4	21.8	14.3	13.4	(19)
(20)			MCWB	8.8	9.0	11.5	12.8	16.3	19.5	20.3	20.4	17.9	15.7	11.2	10.4	(20)
(21)		2%	DB	9.3	11.2	15.2	19.4	24.5	27.5	30.2	29.2	24.9	19.0	12.7	11.2	(21)
(22)			MCWB	7.7	8.2	9.3	11.6	15.7	18.5	19.5	18.7	17.5	14.1	10.8	9.5	(22)
(23)		5%	DB	8.2	8.9	12.5	17.1	22.6	25.3	28.2	27.1	22.4	16.7	11.4	9.7	(23)
(24)			MCWB	6.9	6.8	8.2	10.6	14.7	17.4	18.6	18.0	15.9	13.2	9.9	8.4	(24)
(25)		10%	DB	7.1	7.2	10.7	14.9	20.4	22.9	25.8	24.9	20.1	15.0	10.1	8.0	(25)
(26)			MCWB	6.1	5.7	7.7	9.2	13.7	16.5	17.9	17.0	15.2	12.5	8.7	6.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	10.2	12.0	13.7	18.0	21.3	21.6	20.9	19.2	16.5	12.3	10.8	(27)
(28)			MCDWB	10.4	12.9	16.6	20.4	24.2	27.2	30.1	29.1	24.2	20.6	13.1	11.7	(28)
(29)		2%	WB	8.1	8.6	10.1	12.3	16.4	19.7	20.4	19.6	18.1	15.0	11.2	9.7	(29)
(30)			MCDWB	9.0	10.6	13.6	17.8	23.0	24.7	28.2	27.0	22.6	17.6	12.3	11.2	(30)
(31)		5%	WB	7.1	7.1	9.0	11.1	15.3	18.5	19.3	18.7	17.0	13.7	10.0	8.4	(31)
(32)			MCDWB	7.9	8.6	11.9	15.8	21.3	23.2	26.1	25.2	21.0	16.0	11.2	9.6	(32)
(33)		10%	WB	6.0	5.8	8.0	10.0	14.2	17.3	18.4	17.9	15.9	12.7	8.7	7.0	(33)
(34)			MCDWB	7.1	7.2	10.4	14.0	19.4	21.8	24.4	23.5	19.2	14.6	9.8	7.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.1	6.1	7.4	9.1	9.9	9.5	10.7	10.6	9.2	7.0	4.9	4.0	(35)
(36)			MCDBR	4.9	8.5	11.1	13.1	13.8	13.4	14.7	14.6	13.3	9.8	6.2	5.5	(36)
(37)			MCWBR	4.5	6.0	6.3	6.8	7.1	6.1	6.3	6.0	6.4	5.6	5.0	4.8	(37)
(38)		5% WB	MCDWR	4.9	7.6	9.2	11.3	12.5	11.8	13.1	12.8	11.3	8.2	5.7	5.2	(38)
(39)			MCWBR	4.6	6.0	5.8	6.5	6.7	5.9	5.9	5.9	6.0	5.0	4.9	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.333	0.362	0.392	0.401	0.404	0.401	0.397	0.370	0.355	0.327	0.304	(40)	
(41)		taud	2.435	2.386	2.323	2.274	2.276	2.292	2.292	2.330	2.386	2.424	2.438	2.428	(41)	
(42)		Ebn at Noon	745	807	840	852	858	856	852	837	824	769	707	690	(42)	
(43)		Edh at Noon	62	83	105	122	127	126	124	114	98	79	62	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.80	1.52	2.70	4.24	5.01	5.57	5.29	4.53	3.39	1.96	0.90	0.62	(44)	
(45)		RadStd	0.12	0.29	0.41	0.53	0.51	0.50	0.56	0.46	0.41	0.19	0.10	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page